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DialogSpan: Dialog Correlation for Multi-Agent Systems

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Problem Description

Multi-agent systems and chatbots often answer in isolation. A user may start a thread with one agent, continue with another, and never see whether the two replies agree, contradict, or depend on each other. There is no standard way to correlate dialog turns across agents or to mark when evidence is missing [1].

Key issues:

- NextAgents reply independently. Cross-agent agreement is not checked.
- Contradiction between two agents is invisible to the user.
- There is no shared timeline or evidence link for a multi-agent conversation.
- When correlation cannot be proven, the correct output is Inconclusive, not a forced match.



Background

In multi-agent systems, a task is split across specialized agents. One agent may fetch data, another may reason, and a third may reply to the user. Without correlation, the user cannot see whether the agents used the same facts or contradicted each other [1].

Maya ID builds a correlation layer. It links turns by shared entities, timestamps, and claim overlap. It does not invent a link. If the evidence is missing or contradictory, the result is Inconclusive and no merge is offered [1].

Literature & RFC Review

Multi-agent dialog research treats correlation as a linking problem: which turn depends on which prior turn, and which claims agree. Existing work on dialog state tracking and multi-agent orchestration focuses on routing, not on proving that two agent replies are consistent [1].

Consumer chat interfaces show one reply at a time. They do not surface cross-agent contradiction or missing evidence. Maya ID uses explicit correlation rules and requires a shared evidence span before it marks two turns as linked [2].

Proposed Solution

DialogSpan is a dialog correlation layer for multi-agent systems. It records each agent turn with its source, timestamp, and claimed entities. A deterministic correlation engine then links or rejects pairs.

- Shared entity + overlapping claim → correlated
- Same entity, opposing claim → contradiction
- No shared evidence span → inconclusive
- User may accept a link only after the evidence is shown

No link is created without a supporting span. The user sees the evidence either way.

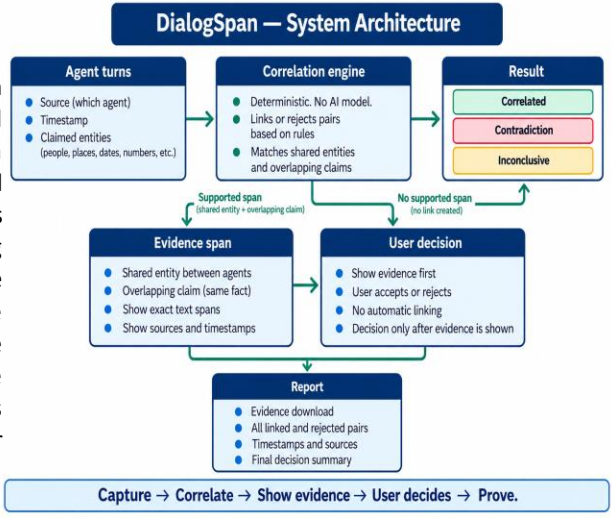
Optimization Proposed

Optimization Proposed — DialogSpan

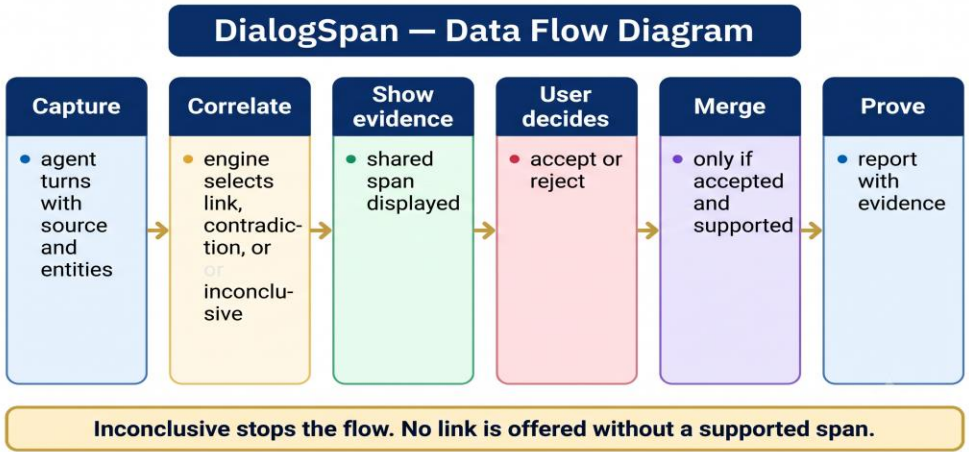
Scenario	Before (Traditional / Current Approach)	After DialogSpan (Proposed Solution)	Benefit
Isolated replies	One agent answer	Linked turns with evidence	User sees agreement
Contradiction	Hidden conflict	Marked and shown	No silent mismatch
Weak evidence	Forced link	Inconclusive. No merge	No false correlation
Demo	Live multi-agent chaos	Lab mode injects known pairs	Repeatable demo
Status	Unknown	Correlated / Contradiction / Inconclusive	Clear outcome

Solution Architecture & Design

DialogSpan records each agent turn with its source, timestamp, and claimed entities. The correlation engine compares pairs using shared entities and claim overlap. A link is created only when a supporting evidence span exists. If the evidence is missing or contradictory, the result is Inconclusive and no merge is offered. The report stores the evidence either way. The user sees the span before accepting or rejecting a link.



Data Flow Diagram



Deployment Timeline



Bibliography I

- [1] Multi-agent dialog systems and dialog state tracking literature. Correlation treated as a linking problem between turns.
- [2] Evidence-based claim linking. A link is valid only when a shared span supports it.
- 3] AIORI. “AIORI-3 Online Hackathon Guidelines.” portal.aiori.in/aiori-3/hackathon-guideline. 2026.



Thank You for Your Attention!